

PROBABILITY

DPP - 1

- Q.1 6 married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is :
- (A) $\frac{16}{33}$ (B) $\frac{8}{33}$ (C) $\frac{17}{33}$ (D) $\frac{24}{33}$
- Q.2 A committee of 5 is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is :
- (A) $\frac{1}{2}$ (B) $\frac{5}{9}$ (C) $\frac{4}{9}$ (D) $\frac{2}{3}$
- Q.3 A quadratic equation is chosen from the set of all the quadratic equations which are unchanged by squaring their roots. The chance that the chosen equation has equal roots is :
- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$
- Q.4 The probability that a positive two digit number selected at random has its tens digit at least three more than its unit digit is
- (A) $\frac{14}{45}$ (B) $\frac{7}{45}$ (C) $\frac{36}{45}$ (D) $\frac{1}{6}$
- Q.5 A 5 digit number is formed by using the digits 0, 1, 2, 3, 4 & 5 without repetition. The probability that the number is divisible by 6 is :
- (A) 8 % (B) 17 % (C) 18 % (D) 36 %
- Q.6 A card is drawn at random from a well shuffled deck of cards. Find the probability that the card is a
- (i) king or a red card (ii) club or a diamond (iii) king or a queen
(iv) king or an ace (v) spade or a club (vi) neither a heart nor a king.
- Q.7 A bag contain 5 white, 7 black, and 4 red balls, find the chance that three balls drawn at random are all white.
- Q.8 If four coins are tossed, Two events A and B are defined as
A: No two consecutive heads occur
B: At least two consecutive heads occur.
Find P(A) and P(B). State whether the events are equally likely, mutually exclusive and exhaustive.
- Q.9 Thirteen persons take their places at a round table, Find the odds against two particular persons sitting together.
- Q.10 A has 3 shares in a lottery containing 3 prizes and 9 blanks, B has 2 shares in a lottery containing 2 prizes and 6 blanks. Compare their chances of success.
- Q.11 There are three works, one consisting of 3 volumes, one of 4 and the other of one volume. They are placed on a shelf at random, find the chance that volumes of the same works are all together.

- Q.12 5 persons entered the lift cabin on the ground floor of an 8 floor building. Suppose that each of them independently and with equal probability, can leave the cabin at any other floor, starting from the first, find the probability that all 5 persons leave at different floors.
- Q.13 Consider a function $f(x)$ that has zeroes 4 and 9. Given that Mr. A randomly selects a number from the set $\{-10, -9, -8, \dots, 8, 9, 10\}$, what is the probability that Mr. A chooses a zero of $f(x^2)$?
- Q.14(a) A fair die is tossed. If the number is odd, find the probability that it is prime.
- (b) Three fair coins are tossed. If both heads and tails appear, determine the probability that exactly one head appears.
- Q.15 n different books ($n \geq 3$) are put at random in a shelf. Among these books there is a particular book 'A' and a particular book B. The probability that there are exactly 'r' books between A and B is
- (A) $\frac{2}{n(n-1)}$ (B) $\frac{2(n-r-1)}{n(n-1)}$ (C) $\frac{2(n-r-2)}{n(n-1)}$ (D) $\frac{(n-r)}{n(n-1)}$
- Q.16 A coin is biased so that heads is three times as likely to appear as tails. Find $P(H)$ and $P(T)$. If such a coin is tossed twice find the probability that head occurs at least once.
- Q.17 Nine number 1, 2, 3,, 9 are put into a 3×3 array so that each number occur exactly once. Find the probability that the sum of the numbers in atleast one horizontal row is greater than 21.
- Q.18 Mr. A lives at origin on the cartesian plane and has his office at (4, 5). His friend lives at (2, 3) on the same plane. Mr. A can go to his office travelling one block at a time either in the + y or + x direction. If all possible paths are equally likely then the probability that Mr. A passed his friends house is
- (A) $1/2$ (B) $10/21$ (C) $1/4$ (D) $11/21$
- Q.19 In a hand at "whist" what is the chance that the 4 kings are held by a specified player?
- Q.20 I have 3 normal dice, one red, one blue and one green and I roll all three simultaneously. Let P be the probability that the sum of the numbers on the red and blue dice is equal to the number on the green die. If P is written in lowest terms as a/b then the value of $(a + b)$ equals
- (A) 79 (B) 77 (C) 61 (D) 57